The U-Shaped Challenge of Language Grounding: How Instruction Granularity Shapes Instruction Following Agent Behavior

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Abstract

Instruction granularity remains an uncontrolled variable in language-guided embodied AI, leaving its impact unknown. We propose an agent-centric formalization of granularity as rule width (w), a cross-task metric adapted from width-based planning. By measuring the peak concurrent state-feature dependencies an agent must resolve, we introduce w as a measure of instruction granularity and show that it consistently correlates with the decision-making burden neural agents face when grounding language. We also introduce MINI-BEHAVIOR-GRAN, a controlled benchmark systematically varying w across 20 tasks while holding perception and control constant. Our analysis reveals a non-monotonic, U-shaped performance curve: while fine-grained instructions ($w \leq 1$) enforce deep semantic alignment, coarse ones ($w \geq 2$) induce shallow grounding and visuomotor shortcuts driven by high instruction predictability ($P(l|v)$). We further uncover a generalization asymmetry, where coarse-trained models transfer to fine instructions but not vice-versa, and show that mixed-granularity training is essential for robust language grounding.

1 Introduction

Despite the rapid progress in language-guided embodied AI, the community has yet to establish a consensus on a unified metric to quantify the granularity of instructions during language grounding. In this domain, language serves a specific functional role: it provides structural information to decompose complex tasks and reduce the search space of the underlying sequential decision-making process. Consequently, *instruction granularity* should not be conflated with superficial syntactic properties, such as sentence length or the number of entities mentioned; rather, it must rigorously reflect the *decision-making burden* that remains for the agent to resolve internally after grounding the language into the action space.

Existing research conceptualizes instruction granularity through two dominant lenses, both of which struggle to provide a consistent measure of the decision-making burden required for grounding. Studies in Natural Language Processing (NLP) typically operationalize granularity via surface-form properties, such as textual specificity or discourse structure (e.g., number of entities mentioned) (Dong and Lapata, 2018; Huang et al., 2025b). While these effectively measure syntactic complexity, they adopt a *text-centric* rather than *agent-centric* view. Consequently, they cannot quantify how much an instruction reduces an embodied agent's *decision-making burden*.

From an *agent-centric* perspective, the essential function of instruction is to decompose complex tasks into tractable sub-problems, thereby reducing the planning burden (Drexler et al., 2022). Recent work in LLM-based agents thus treats granularity as *decomposition depth*—the extent to which an instruction enumerates subgoals versus stating only the final objective (Yang et al., 2024; Myers et al., 2024; Shi et al., 2025; Zhong et al., 2025). While this aligns with our view that *instruction granularity* modulates planning demand, decomposition depth remains an unstable, task-dependent proxy. It conflates instructional guidance with the task's inherent structure: complex tasks with interleaved subgoals may resist breakdown (e.g., the Sussman Anomaly (Sussman, 1975)), while trivial ones can be artificially elongated.

To operationalize this, we draw inspiration from the concept of policy sketches (Drexler et al., 2024) and formalize instruction granularity as rule width (w) (Bonet et al., 2019). We hypothesize that w captures the decision-making burden that an instruction imposes, and we provide initial empirical evidence supporting this hypothesis, showing that it is a more reliable indicator than common linguistic heuristics. To this end, the contributions of our paper are:

(1) Width as a Formalism for Instruction Granularity: We formalize instruction granularity as *rule width* (w), adapting a concept from width-based planning (Bonet et al., 2019; Drexler et al., 2024) that measures the peak number of concurrent state variables (features Φ) that must be tracked to fulfill the instructions. We argue w measures the *intrinsic decision complexity* of a task-instruction pair, applicable to both symbolic and neural agents (§ 3).

(2) A Controlled Benchmark: We present MINI-BEHAVIOR-GRAN, the first testbed to isolate instruction granularity from confounding factors like task horizon and perception. It enables the first study of how instruction granularity affects language-guided embodied agent performance (§ 4).

(3) The U-Shaped Trend and Width Cap: In this controlled benchmark, we identify a non-monotonic U-shaped relationship where performance peaks at the Fine ($w = 0$) and Coarse ($w \geq 2$) extremes. We also identify a performance collapse at high width ($w \geq 4$): agents fail to fulfill instructions requiring maintaining concurrent features ≥ 4, revealing a sharp language grounding boundary within our experimental setting (§ 5).

(4) Shallow Grounding and Visuo-Motor Shortcuts: We diagnose the performance rebound at $w \geq 2$ not as true mastery, but as a shift toward *shallow grounding*. Driven by high instruction predictability ($P(l|v)$), agents increasingly rely on visuo-motor shortcuts rather than deep semantic alignment. Through controlled semantic perturbations, we show that mixed-granularity training mitigates this vulnerability, equipping agents with deeper semantic resilience against suboptimal instructions. Additionally, we identify an *asymmetric generalization* pattern: agents trained on coarse instructions successfully transfer to fine-grained ones, but not vice versa.

2 Related Work

Instruction Abstraction and Granularity. In this work, *Instruction granularity* refers to the extent to which state and action constraints are explicitly specified versus left for the agent to infer when grounding language into actions. This is distinct from, yet often conflated with the concept of *linguistic abstraction*. The term *abstraction* is viewed as *representational compression*—using a compact description to stand for multiple instances or executions. This definition is common in prior work of computational thinking (Wing, 2010; Lachmy et al., 2022; Zheng et al., 2024). Importantly, granularity can vary independently of abstraction: “tidy the room” is coarse but non-abstract (it leaves the exact required actions underspecified, yet it lacks representational compression.), whereas “repeat wiping until clean” is relatively specific yet abstract (it prescribes an action but uses a loop condition to compress multiple executions).

While not framed as the term “granularity”, we acknowledge that a rich body of work has focused on augmenting language specificity and its impact, for example through coarse-to-fine semantic parsing (Dong and Lapata, 2018; Li et al., 2020). Other work generates compositional instructions via constraint dropout and sub-task recombination (Huang et al., 2025b), or construct counterfactual instructions to augment data diversity (Glossop et al., 2025).

Varying instruction detail is often handled via a simple two-part decomposition: high-level plans are converted into sequences of low-level steps before being grounded into actions (Yang et al., 2024; Shi et al., 2025). Most recent Vision Language Action (VLA) works, however, operate with static language instructions (Ma et al., 2025; Li et al., 2025). Some models like InstructVLA and ThinkAct explore having chain-of-thought reasoning steps before action decoding (Yang et al., 2025; Huang et al., 2025a). These hierarchical definitions remain task-specific and qualitative, lacking a unified, cross-task metric to establish shared granularity levels across tasks. Consequently, quantitatively controlled studies on the formal dynamics of instruction granularity remain scarce in the embodied AI domain.

Language Grounding vs. Shortcuts. Prior work highlights a dichotomy between fragile language sensitivity (Akula et al., 2022) and visuo-motor shortcuts (Xing et al., 2025). We reconcile this via rule width (w): low w exhibits low predictability ($P(l|v)$), enforcing deep language grounding. Conversely, $w \geq 2$ yields high $P(l|v)$: this redundancy induces a shift toward shallow grounding, where agents default to visual priors.

3 Rule Width as Instruction Granularity

3.1 Running Example

We first illustrate *planning width* computation using a running example. Table 1 shows the desired

State Progression	Concurrent Features Tracked (Sequence of Tuples)	Width
Initial State	$[\langle\rangle]$	0
Navigate to rag	$[\langle p_r = X\rangle, \ldots, \langle p_r = 0\rangle]$	1
Pick up rag	$[\langle p_r = 0\rangle, \langle H_r\rangle]$	1
Navigate to sink	$[\langle H_r, p_s = X\rangle, \ldots, \langle H_r, p_s = 0\rangle]$	2
Turn on tap (sink)	$[\langle H_r, p_s = 0\rangle, \langle T_s\rangle]$	2
Soak rag	$[\langle H_r, T_s, p_s = 0\rangle, \langle S_r\rangle]$	**3**
Navigate to Shoe 1	$[\langle H_r, S_r, p_{f1} \downarrow\rangle, \ldots,]$	**3**
Clean Shoe 1	$[\langle H_r, S_r, p_{f1} = 0\rangle, \langle C_1\rangle]$	3
Navigate to Shoe 2	$[\langle H_r, S_r, C_1, p_{f2} \downarrow\rangle, \ldots]$	**4**
Clean Shoe 2	$[\langle H_r, S_r, C_1, p_{f2} = 0\rangle, \langle C_1, C_2\rangle]$	4
Put rag on floor	$[\langle C_1, C_2, H_r\rangle, \ldots, \langle C_1, C_2, \text{onfloor}_r\rangle]$	3

Table 1: State progression and width for the "Cleaning k Shoes" ($k = 2$). Width scales as $w = k + 2$.

state progressions of a task where two shoes are cleaned using a rag, and the rag is placed on the floor afterwards. The key **environment dynamics** are: (1) the rag must be soaked before it can clean shoes, and (2) the tap (sink) must be turned on to soak the rag. We represent states using a set of task-relevant feature variables (Φ): (i) boolean features H_r (holding rag), T_s (tap (sink) is turned on), S_r (rag is soaked), C_i (shoe i is cleaned), $onfloor_r$ (rag is on the floor); and (ii) spatial features p_r, p_s, p_f (distance to the rag, sink, and nearest uncleaned shoe, respectively).

First, approaching the rag is a navigation-only segment: the desired state progression is to decrease the distance feature p_r until $p_r = 0$. Thus, the width is 1 as only a single feature needs to be tracked. In this environment dynamics, $p_r = 0$ enables the state transition to H_r. Thus, the rag can be obtained at $p_r = 0$ and the width remains 1. Subsequently, H_r must be maintained while progressing to next states, resulting in a width of 2. The rag becoming soaked requires simultaneous maintenance of three features, i.e., at the sink ($p_s = 0$), holding the rag (H_r), and having the tap (sink) turned on (T_s), which yields a width of 3. After soaking, C_1 can be obtained by tracking H_r, S_r, and p_f, giving width 3. The subtlety arises when cleaning the second shoe: one has to track the same features as before, plus maintaining the cleanliness of the first shoe (C_1), which increases the width to 4. A special case of $w = 0$ occurs for situations where the desired state progression can be achieved in zero or one step (Bonet and Geffner, 2024); this will appear in our experimental analysis.

Why no action model? Agents' action sets constrain how state transitions are executed, but *width* focuses on state-level feature dependencies induced by the environment dynamics. Different agents (e.g., a rover vs. a drone) may have different action spaces to achieve the same feature change, so emphasizing state progressions promotes generalization of width-based concepts across different embodiments.

The impact of *instruction granularity* on language-guided embodied agent performance remains underexplored largely because the field lacks a **formal, quantifiable, cross-task** measure of granularity. Building on the limitations of heuristic metrics discussed in § 1, we formally equate *instruction granularity* with *latent planning demand*, i.e., the implicit decision-making burden required to ground text into executable actions. We operationalize this using *rule width (w)* by framing natural-language instructions as policy sketches (Drexler et al., 2024). Under this framework, fine-grained instructions decompose tasks into minimal-width sub-problems, whereas coarse instructions leave a large "grounding gap" that the agent must resolve internally.

3.2 Symbolic Task Representation

While our agents will be trained on raw multimodal observations, we adopt a symbolic, relational state abstraction. Many embodied benchmarks provide such structured representations—object properties and relations—that can be expressed as *fluents* to form symbolic state or goal conditions (Shridhar et al., 2020; Li et al., 2024). Formally , we consider a classical planning problem $P = \langle \mathcal{F}, \mathcal{A}, I, G\rangle$, where $\mathcal{F}$ is a set of fluents, $\mathcal{A}$ a set of actions, I the initial state, and G the goal.

To bridge raw states and high-level instructions, we define a set of **features** $\Phi = \{\phi_1, \ldots, \phi_n\}$, where ϕ_i is a Boolean or integer-valued function over state space S. These features serve as domain-level abstractions that capture task-relevant aspects of the environment. Thus, each s is associated with a **feature valuation vector** $\phi(s) = (\phi_1(s), \ldots, \phi_n(s))$.

3.3 Instructions as Rule Sets $\mathcal{R}$

We formalize each instruction as a symbolic interpretation, represented as a set of "if-then" rules $\mathcal{R} = \{r_1, \ldots, r_m\}$ defined over the feature space Φ, This rule-based representation does not aim to capture the full expressiveness of natural language; instead, it distills the constraint/goal structure that governs the computational cost incurred by a symbolic search algorithm (i.e., planning demand). Specifically, each r_i is of the form $C_{r_i} \mapsto E_{r_i}$:

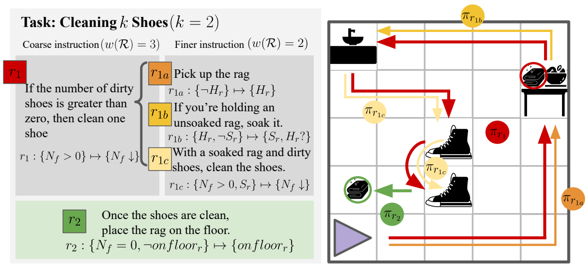

Figure 1: (Left) Factoring r_1 into r_{1a}-r_{1c} reduces latent planning complexity via explicit action guidance. (Right) Fine instructions segment the long-horizon coarse trajectory (π_{r_1}, red).

- C_{r_i} (the *condition*) is a conjunction of feature constraints. For a Boolean feature p, a constraint is p or $\neg p$; for a numerical feature n, it is $n = 0$ or $n > 0$.
- E_{r_i} (the *effect*) is a conjunction of feature changes. For a Boolean feature p, an effect is p, $\neg p$, or $p?$ (allowing any change); for a numerical feature n, it is $n \downarrow$ (decrease), $n \uparrow$ (increase), or $n?$ (any change).

These forms capture the core semantics of the instructions we study. A rule r_i is **fulfilled** by a state sequence $(s, \ldots, s')$ if $\phi(s) \models C_{r_i}$ and the pair $(\phi(s), \phi(s'))$ satisfies the changes in E_{r_i} (e.g., $\forall (n \downarrow) \in E_{r_i}, \phi_n(s') < \phi_n(s)$) while keeping unmentioned features constant (i.e., $\phi_j(s') = \phi_j(s)$ for $j \notin E_{r_i}$). To ensure that the agent's performance strictly reflects its grounding ability rather than the unsolvability of the task, we guarantee that all instruction rule sets $\mathcal{R}$ are executable and dead-end-free. Formally, this is achieved by ensuring the rules are well-formed (i.e., they terminate at the goal) and satisfy the soundness and closure properties (Bonet and Geffner, 2021).

3.4 Rule Width as a Granularity Measure

The width $w(r_i)$ of a rule r_i represents the number of state variables (features) that must be tracked concurrently to fulfill the transition. Formally, for a rule to have width k, the agent must coordinate at most k features from Φ to reach the effect E_{r_i} from a state satisfying C_{r_i} (Lipovetzky and Geffner, 2012).

We define the granularity of an instruction $\mathcal{R}$ by its width-based planning bottleneck: the maximum width among its constituent rules: $w(\mathcal{R}) = \max_{r_i \in \mathcal{R}} w(r_i)$.

We continue the running example to demonstrate two instruction sets, each representing a different width level, with two variants visualized in Figure 1. We also append a **counting feature** in Φ to track the number of uncleaned shoes, denoted as N_f.

Coarse-Grained Instruction ($w(\mathcal{R}) = 3$) This instruction states only the high-level rules and their desired ordering, leaving the internal structure of each sub-task implicit:

- $r_1 : \{N_f > 0\} \mapsto \{N_f \downarrow\}$ (clean a shoe; $w = 3$)
- $r_2 : \{N_f = 0, \neg onfloor_r\} \mapsto \{onfloor_r\}$ (put the rag on the floor after cleaning shoes; $w = 1$)

Middle-Grained Instruction ($w(\mathcal{R}) = 2$) This variant decomposes r_1 to explicitly state the *soaking* process:

- $r_{1a} : \{\neg H_r\} \mapsto \{H_r\}$ (pick up the rag; $w = 1$)
- $r_{1b} : \{H_r, \neg S_r\} \mapsto \{S_r, H_r?\}$ (soak the rag)
- $r_{1c} : \{N_f > 0, S_r\} \mapsto \{N_f \downarrow\}$ (clean; $w = 2$)
- $r_2 : \{N_f = 0, \neg onfloor_r\} \mapsto \{onfloor_r\}$ (put the rag on the floor; $w = 1$)

This rule set is well-formed: it is goal-terminating via r_2, and the rules are sequentially reachable, meaning the successor state of any fulfilled rule (that is not the goal) also meets the condition of at least one rule in $\mathcal{R}$. The set is also robust to perturbations: if an agent drops the rag after soaking it ($\neg H_r$), the rule set remains applicable by reverting to r_{1a} to reacquire the rag before proceeding to the cleaning task (r_{1c}).

Width as a Planning Proxy. Decomposing a task into a rule set $\mathcal{R}$ provides structural scaffolding, guiding the agent through intermediate "stepping stone" states. By design, we assume the execution of r_i does not negate established fluents unless explicitly required by E_{r_i}. This is the essence of language-guided decomposition. Without this assumption, language guidance would offer no reduction in complexity; the agent would need to track every feature since the initial state.

This intuition is formally grounded in novelty-driven Breadth-First Search (Lipovetzky and Geffner, 2017). By pruning states that fail to achieve a new feature conjunction of size $\leq k$, search complexity is bounded by $O(|\Phi|^k)$. Thus, trajectories that unnecessarily negate an already-achieved precondition C_{r_i} (unless required by E_{r_i}) are longer than those preserving it and are pruned from the optimal frontier. Consequently, features

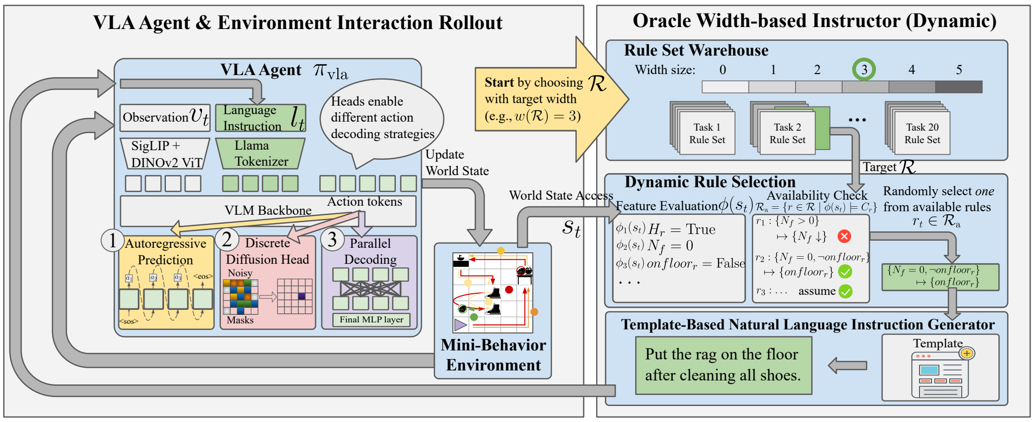

Figure 2: MINI-BEHAVIOR-GRAN Experimental Pipeline. We evaluate the impact of instruction granularity across three VLA action-decoding strategies. During inference time, an Oracle Instructor evaluates the current world state to decide available rule set $\mathcal{R}_a$ and active rule r_t. A rule-based natural language instruction l_t is generated accordingly. The process iterates until completion or max steps.

in C_{r_i} that are already true are effectively treated as invariants, and the active coordination w reduces to the incremental features needed for the next subgoal. For instance, while the monolithic task requires coordinating $\{H_r, p_s, T_s\}$ ($w = 3$, Table 2), the instruction r_{1b} leverages the fact that H_r is pre-established. By treating H_r as a fixed invariant, the search width w reduces to the novel features $\{p_s, T_s\}$, yielding $w = 2$.

To communicate these rules to language-guided embodied agents, we translate each rule into a natural-language sentence via a template-based generator. These templates verbalize the features within each rule; e.g., $\{\ H_r\ ,\ \neg S_r\ \} \mapsto \{\ S_r\ \}$ is realized as "If you are holding the rag but the rag is not yet soaked, soak the rag." This construction preserves the granularity differences defined by the rule width. Additional language prompt examples are provided in § F.

4 Mini-BEHAVIOR-Gran Benchmark

To disentangle the influence of instruction granularity in agent performance, we seek an environment that (1) retains long-horizon and heterogeneous activities so that instruction guidance meaningfully modulates the planning demand, while (2) simplifies perception and low-level control to avoid confounding factors. As such, we use MINI-BEHAVIOR (Jin et al., 2023) as our testbed. Detailed justification for our choice over other environments is provided in § B.

We introduce MINI-BEHAVIOR-GRAN, the first benchmark enabling controlled study of instruction granularity via *rule width*. Over 20 tasks sharing a feature set Φ, we design rule sets at up to six width levels, all verified solvable. During inference, an oracle instructor monitors the current feature vector $\phi(s_t)$, selects an available rule r_t from the task's rule set, and translates it via a template into a natural language instruction l_t. This guarantees that performance differences reflect grounding ability alone, not stochastic instruction quality (see Figure 2). The instruction l_t and current observation v_t are then fed to the learning-based agent to predict action chunks $(a_t, ..., a_{t+k}) \sim \pi_{\text{agent}}(\cdot \mid v_t, l_t)$.

5 Experiments

5.1 Experimental Setup

Task Suite & Training. We use MINI-BEHAVIOR-GRAN, which contains 20 complex tasks sharing the same feature set Φ. We further classify instruction granularity into three *relative* classes of *Fine (F)*, *Medium (M)*, and *Coarse (C)* via per-task quantile binning. In most cases, this mapping corresponds to $w = 0$ for Fine, $w = 1$ for Medium, and $w \geq 2$ for Coarse. We evaluate seven training settings: (1) **Pure F/M/C** $(1:0:0)$: trains exclusively on one granularity class; (2) **Centroid** $(1:1:1)$: uniformly sample from all three classes; (3) **Axial F/M/C** $(3:1:1)$: emphasizes one class with some exposure to the other classes. Our dataset provides 50 training and 10 evaluation instances per task, each paired with reference trajectories generated by BFWS width-based planner from Lipovetzky and Geffner (2017) (see § E for details).

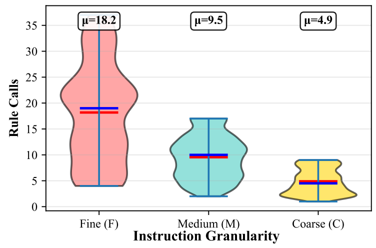

Figure 3: Rule call distribution across instruction granularities. # of rule calls decreases with coarser instructions. It decouples instruction length from width.

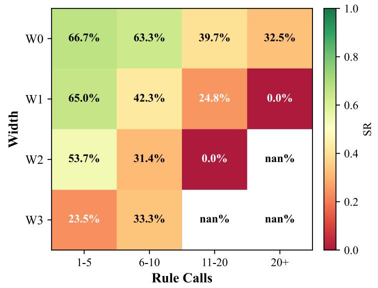

Figure 4: Impact of width and horizon on SR. SR decays with task horizon; larger instruction width imposes an additional burden even when horizon is fixed.

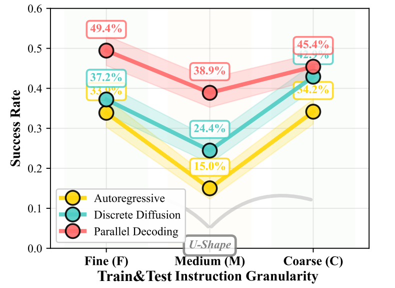

Figure 5: In-Distribution (matched train/test) SR across granularities (i.e., train Fine → test Fine, etc). A non-monotonic U-shaped trend appears.

Action-Decoding Archetypes. The primary architectural difference among latest VLA models lies in their action-decoding strategies (Zhong et al., 2025). Hence, we employ a unified VLM backbone but vary three representative action-decoding strategies (see Figure 2 left panel): (1) **Autoregressive** (AR), exampled by RT-2 (Zitkovich et al., 2023), OpenVLA (Kim et al., 2024) and others (Reed et al., 2022; Huang et al., 2024; O'Neill et al., 2024) that predict actions sequentially via causal attention; (2) **(Discrete) Diffusion**, which uses diffusion process for action chunk prediction (Black et al., 2025; Bjorck et al., 2025; Shukor et al., 2025). We employ discrete diffusion (Liang et al., 2025) for compatibility with the discrete action space of our testbed; (3) **Parallel Decoding**, utilizing bidirectional attention for single-pass action prediction to improve throughput, as demonstrated by OpenVLA-OFT (Kim et al., 2025b), PD-VLA (Song et al., 2025) and other variants (Yin et al., 2025; Wang et al., 2025; Xiao et al., 2025). Details on model hyperparameters, hardware and training schedules are in § G.

5.2 Establishing the Yardstick: Decoupling Horizon from Width

A critical confounder in evaluating language grounding complexity is the task sequence length. However, in language-guided execution, instructions inherently decompose long-horizon tasks into segments. Consequently, the relevant measure of sequential burden shifts from the raw atomic horizon (total action steps) to the instruction horizon, i.e., the number of rule calls issued by the instructor.

To isolate rule width from sequential burden, we train a single agent across all tasks using the Centroid configuration, ensuring balanced exposure across all granularity levels. This approach holds the distribution of atomic horizons constant, allowing us to isolate how instruction granularity and the frequency of rule calls affect performance. Figure 3 shows that fine-grained instructions ($w = 0$) decompose the task into numerous low-width rule calls; coarse instructions ($w \geq 2$), by contrast, yield fewer rule calls.

To assess whether w serves as an effective indicator of latent planning demand for learning-based agents, we analyze Figure 4. The heatmap reveals a strict monotonic decay in Success Rate (SR) as rule width increases. Also, for a fixed w, SR decays as rule calls increase, mirroring greater sequential horizon complexity. This confirms w as a horizon-independent measure of the intrinsic planning demand imposed by instruction granularity.

Width vs. Syntactic Heuristics. We evaluate whether w captures language grounding complexity better than common linguistic heuristics: token count, entity count, and action verbs. Figure 8 shows w is the only metric that exhibits a statistically significant and consistently *negative* correlation with SR across task horizons. Action Verbs correlate significantly but flip direction across horizons, while Token Count turns negative only in the longest bin (20+ steps). These heuristics are inconsistent compared to w (see details in § H).

5.3 The U-Shaped Paradox and Capacity Cap

To establish a baseline, we train models using the *Pure F/M/C* setting. the models are trained and evaluated on the same granularity class. As shown in Figure 5, all VLA archetypes exhibit a non-monotonic *U-shaped performance curve*.

At the fine extreme ($w = 0$), where instructions

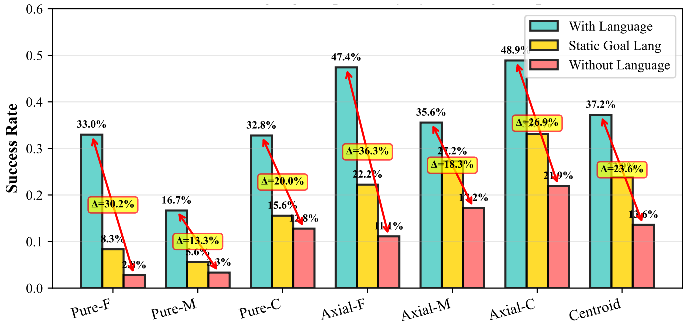

Figure 6: Smaller SR gaps indicate a shift towards shallow grounding. Under semantic perturbation, the Pure-C setting exhibits tiny variance (2.8%) between *w/ goal lang* and *w/o lang*.

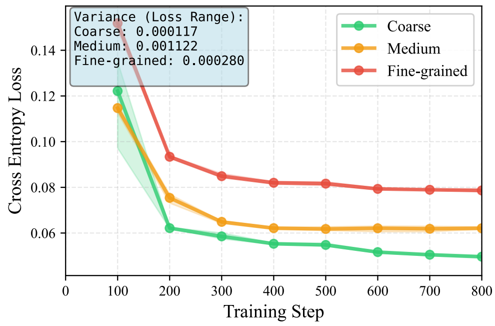

Figure 7: VLM instruction-prediction Cross Entropy (CE) loss across granularities and 3 seeds. Larger CE loss indicates lower $P(l \mid v)$.

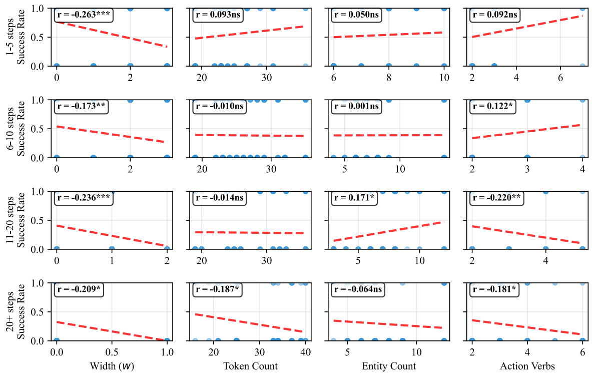

Figure 8: Metric-success correlation by task-horizon.

require minimal latent reasoning to be grounded into actions, models achieve high performance. Performance then decreases at medium granularity ($w = 1$), before unexpectedly rebounding at the coarse regime ($w \geq 2$). This contradicts the assumption that higher planning width should monotonically degrade learning.

5.4 Mechanistic Probe: The Grounding Valley

Why does performance rebound at $w \geq 2$? We show this arises from increasing *instruction predictability* $P(l|v)$, which enables agents to bypass language and exploit visual-dominant policy.

Following prior work (Fei et al., 2025), we use $\Delta\mathrm{SR}_{\text{w/ - w/o}}$ as a basic measure of language grounding. To further isolate grounding effects, we conduct another intervention: replacing full instructions with high-level static goal descriptions (Figure 6). When language is entirely removed, fine-trained policies suffer a severe 36.3% drop, whereas coarse-trained policies show a smaller 20.0% gap. Also note that under reduced informativeness (*w/ goal lang*), the Pure-C policy exhibits a negligible 2.8% advantage over having no language at all (15.6% vs. 12.8%). This indicates that policies trained on coarse instructions develop a shallow grounding behavior: language is not deeply integrated into their decision-making. Consequently, when instructions are reduced to generic goals, their performance barely differs from the vision-only baseline.

The Predictability Driver: Why Reliance Shifts. We hypothesize that this behavioral shift originates from coarse instructions being statistically more predictable from vision. Formally, consider a surjective mapping f from fine-grained labels L_{fine} to coarser ones L_{coarse}. The conditional probability of a coarse label aggregates its fine-grained constituents: $P(c_j|v) = \sum_{l_i \in f^{-1}(c_j)} P(l_i|v)$. The increasing predictability of coarse instructions reduces the marginal information contributed by language, making it easier for agents to minimize task loss by relying primarily on visual observations, thereby favoring the development of visuomotor shortcuts over costly cross-modal alignment. We confirm this predictability empirically by training a VLM (Qwen3-VL-4B via LoRA) to act as an trained instructor, predicting instructions from visual observations. As Figure 7 shows, Cross-Entropy (CE) loss decreases monotonically with coarseness (Fine: 0.078, Medium: 0.063, Coarse: 0.047), a relative drop of 40% from Fine to Coarse. Together, these results form a complementary chain: probability aggregation explains *why* $P(l|v)$ increases, CE loss empirically confirms the increase, and the perturbation ΔSR quantifies the resulting visuo-motor shift.

Thus, we attribute this U-shaped behavior to how the agent dynamically shifts its reliance on text versus vision across different rule widths:

- **Fine ($w = 0$):** *Deep Grounding.* Low $P(l|v)$ (high CE loss, Figure 7) means language pro-

Training Setup	Width 0	Width 1	Width 2	Width 3	Width ≥ 4
Pure-F	34.9%	16.9%	10.8%	1.0%	0.0%
Pure-M	12.2%	22.0%	11.7%	0.0%	0.0%
Pure-C	26.3%	33.3%	32.9%	22.9%	0.0%
Axial-F	50.2%	53.3%	48.8%	34.3%	0.0%
Axial-M	44.7%	43.5%	45.8%	29.5%	0.0%
Axial-C	**55.7%**	52.2%	50.8%	**37.1%**	0.0%
Centroid (Mix)	42.7%	38.4%	40.8%	28.6%	0.0%

Table 2: The x-axis are test instruction width, y-axis are training setup. Transfer is asymmetric: train coarse $\rightarrow$ test fine works, but train fine $\rightarrow$ test coarse does not. Mixed-granularity improves overall.

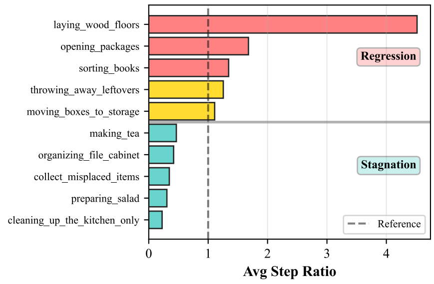

Figure 9: Low-width but long task horizon (e.g., laying woods) tends to cause regression, while high-width stagnates.

vides unique information. Combined with low-width grounding burden (§ 5.2), the model reliably maps instructions to actions, yielding high SR (Figure 5).

- **Medium** ($w = 1$)**:** *The Alignment Trap.* At this width, grounding complexity increases while $P(l|v)$ remains moderate: not yet high enough to exploit reliable visuo-motor shortcuts. The model must attempt language grounding but struggles with its inherent difficulty. This intermediate regime, caught between hard alignment and insufficient shortcuts, yields the lowest performance.
- **Coarse** ($2 \leq w \leq 3$)**:** *Reduced Language Reliance.* High $P(l|v)$ incentivizes a visual-dominant policy: aligning vision to action is easier than learning cross-modal grounding. This yields high in-distribution scores but at the cost of language grounding capability.
- **Capacity Cap** ($w \geq 4$)**:** *Cognitive Overload.* Mirroring the $O(|\Phi|^4)$ complexity barrier of symbolic planners (Lipovetzky and Geffner, 2017), VLA agents performance uniformly collapses at $w \geq 4$ (Table 2). This convergence suggests w as a fundamental representational bottleneck shared by both symbolic and neural paradigms.

Deepening Grounding via Axial Training. Having diagnosed this shallow reliance as a byproduct of high predictability, we hypothesize that increasing the diversity of the training language space L should artificially lower the average $P(l|v)$, forcing the agent to structurally re-engage with the text. Our mixed-granularity training confirms this mechanism. All mixed-granularity models exhibit substantially larger ΔSR gaps compared to their *Pure-X* counterparts. Among these, the fine-weighted mixture *Axial-F* emerges as the optimal structural compromise: it achieves an overall success rate, while maintaining a significantly larger ΔSR.

5.5 Other Results

Asymmetric Generalization. We observe an asymmetric transfer pattern (Table 2): coarse-trained models successfully zero-shot transfer to fine-grained instructions, but fine-trained models fail on coarse-grained ones. This asymmetry aligns with our ΔSR analysis: coarse training induces a visual-dominant policy, hence perturbing or refining the instruction yields minimal behavioral change. Conversely, fine training induces tight coupling with language, making the policy brittle when detailed guidance is withdrawn.

Failure Modes. Agent failure distributions bifurcate strictly with width (Figure 9). Low-width, long-horizon tasks predominantly suffer from *regression*: the agent violates previously satisfied fluents and cycles back to prior states, reflecting subgoal serialization failures. High-width tasks ($w \geq 3$), by contrast, exhibit *stagnation*: the agent freezes, unable to resolve concurrent feature dependencies ($O(|\Phi|^w)$) required to initiate a valid state transition. This dichotomy mirrors the width-based complexity landscape (see § H for details).

6 Conclusion

This work introduces rule width (w) as a cross-task formalism for instruction granularity and demonstrates that it correlates with language grounding difficulty in learning-based agents. We observed a non-monotonic relationship between granularity and performance, driven by a trade-off between instruction predictability ($P(l|v)$) and latent planning complexity. We demonstrate that mixed-granularity training balances language sensitivity without sacrificing the robust visual reasoning skills learned from coarse trajectories.

7 Limitations

Discrete Action Spaces. We evaluate our framework in environments with discrete action spaces, a common abstraction in language-conditioned embodied AI (e.g., ALFWorld-based agent (Xiong et al., 2025; Kim et al., 2025a)) and LLM-based agent systems (Qin et al., 2024; Patil et al., 2024), where high-level reasoning is decoupled from low-level execution via skill primitives or API calls. This choice is intentional: it isolates symbolic planning from continuous control noise, allowing us to cleanly attribute observed effects to instruction granularity. However, extending this granularity-aware framework to hybrid discrete-continuous or continuous action spaces remains essential for real-world robotics applications, where low-level motor commands interact with high-level linguistic abstractions.

On the Oracle Instructor Setup. A potential concern is that providing the active rule via an Oracle simplifies the task to single-step grounding. However, we argue that the **latent planning demand** w is an intrinsic property of the instruction-to-action mapping. Even when the "what to do" is provided, the agent must still coordinate w concurrent features to achieve the transition. Our findings that performance collapses at $w \geq 4$ *despite* Oracle guidance (Table 2) demonstrates that the grounding gap is a fundamental bottleneck in the learning process.

References

Arjun R. Akula, Spandana Gella, Aishwarya Padmakumar, Mahdi Namazifar, Mohit Bansal, Jesse Thomason, and Dilek Hakkani-Tur. 2022. ALFRED-L: investigating the role of language for action learning in interactive visual environments. In *EMNLP*, pages 9369–9378. Association for Computational Linguistics.

Johan Bjorck, Fernando Castañeda, Nikita Cherniadev, Xingye Da, Runyu Ding, Linxi Fan, Yu Fang, Dieter Fox, Fengyuan Hu, Spencer Huang, and 1 others. 2025. Gr00t n1: An open foundation model for generalist humanoid robots. *arXiv preprint arXiv:2503.14734*.

Kevin Black, Noah Brown, James Darpinian, Karan Dhabalia, Danny Driess, Adnan Esmail, Michael Robert Equi, Chelsea Finn, Niccolo Fusai, Manuel Y. Galliker, Dibya Ghosh, Lachy Groom, Karol Hausman, brian ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Devin LeBlanc, Sergey Levine, and 16 others. 2025. $\pi_{0.5}$: a vision-language-action model with open-world generalization. In *Proceedings of The 9th Conference on Robot Learning*, volume 305 of *Proceedings of Machine Learning Research*, pages 17–40. PMLR.

Blai Bonet, Guillem Francès, and Hector Geffner. 2019. Learning features and abstract actions for computing generalized plans. In *AAAI*, pages 2703–2710. AAAI Press.

Blai Bonet and Hector Geffner. 2021. General policies, representations, and planning width. In *AAAI*, pages 11764–11773. AAAI Press.

Blai Bonet and Hector Geffner. 2024. General policies, subgoal structure, and planning width. *Journal of Artificial Intelligence Research*, 80:475–516.

Li Dong and Mirella Lapata. 2018. Coarse-to-fine decoding for neural semantic parsing. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 731–742, Melbourne, Australia. Association for Computational Linguistics.

Dominik Drexler, Jendrik Seipp, and Hector Geffner. 2022. Learning sketches for decomposing planning problems into subproblems of bounded width. In *ICAPS*, pages 62–70. AAAI Press.

Dominik Drexler, Jendrik Seipp, and Hector Geffner. 2024. Expressing and exploiting subgoal structure in classical planning using sketches. *J. Artif. Intell. Res.*, 80.

Senyu Fei, Siyin Wang, Junhao Shi, Zihao Dai, Jikun Cai, Pengfang Qian, Li Ji, Xinzhe He, Shiduo Zhang, Zhaoye Fei, Jinlan Fu, Jingjing Gong, and Xipeng Qiu. 2025. Libero-plus: In-depth robustness analysis of vision-language-action models. *Preprint*, arXiv:2510.13626.

Catherine Glossop, William Chen, Arjun Bhorkar, Dhruv Shah, and Sergey Levine. 2025. Cast: Counterfactual labels improve instruction following in vision-language-action models. *Preprint*, arXiv:2508.13446.

Danijar Hafner. 2021. Benchmarking the spectrum of agent capabilities. *arXiv preprint arXiv:2109.06780*.

Chi-Pin Huang, Yueh-Hua Wu, Min-Hung Chen, Yu-Chiang Frank Wang, and Fu-En Yang. 2025a. Thinkact: Vision-language-action reasoning via reinforced visual latent planning. *Preprint*, arXiv:2507.16815.

Hui Huang, Jiaheng Liu, Yancheng He, Shilong Li, Bing Xu, Conghui Zhu, Muyun Yang, and Tiejun Zhao. 2025b. Musc: Improving complex instruction following with multi-granularity self-contrastive training. In *ACL (1)*, pages 10667–10686. Association for Computational Linguistics.

Jiangyong Huang, Silong Yong, Xiaojian Ma, Xiongkun Linghu, Puhao Li, Yan Wang, Qing Li, Song-Chun Zhu, Baoxiong Jia, and Siyuan Huang. 2024. An embodied generalist agent in 3d world. In *ICML*.

Emily Jin, Jiaheng Hu, Zhuoyi Huang, Ruohan Zhang, Jiajun Wu, Li Fei-Fei, and Roberto Martín-Martín. 2023. Mini-BEHAVIOR: A procedurally generated benchmark for long-horizon decision-making in embodied AI. In *NeurIPS 2023 Workshop on Generalization in Planning*.

Siddharth Karamcheti, Suraj Nair, Ashwin Balakrishna, Percy Liang, Thomas Kollar, and Dorsa Sadigh. 2024. Prismatic vlms: Investigating the design space of visually-conditioned language models. In *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*.

Jeonghye Kim, Sojeong Rhee, Minbeom Kim, Dohyung Kim, Sangmook Lee, Youngchul Sung, and Kyomin Jung. 2025a. ReflAct: World-grounded decision making in LLM agents via goal-state reflection. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 33433–33465, Suzhou, China. Association for Computational Linguistics.

Moo Jin Kim, Chelsea Finn, and Percy Liang. 2025b. Fine-tuning vision-language-action models: Optimizing speed and success. *arXiv preprint arXiv:2502.19645*.

Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan Paul Foster, Pannag R. Sanketi, Quan Vuong, Thomas Kollar, Benjamin Burchfiel, Russ Tedrake, Dorsa Sadigh, Sergey Levine, Percy Liang, and Chelsea Finn. 2024. Openvla: An open-source vision-language-action model. In *CoRL*, volume 270 of *Proceedings of Machine Learning Research*, pages 2679–2713. PMLR.

Royi Lachmy, Valentina Pyatkin, Avshalom Manevich, and Reut Tsarfaty. 2022. Draw me a flower: Processing and grounding abstraction in natural language. *Trans. Assoc. Comput. Linguistics*, 10:1341–1356.

Chengshu Li, Fei Xia, Roberto Martín-Martín, Michael Lingelbach, Sanjana Srivastava, Bokui Shen, Kent Elliott Vainio, Cem Gokmen, Gokul Dharan, Tanish Jain, Andrey Kurenkov, Karen Liu, Hyowon Gweon, Jiajun Wu, Li Fei-Fei, and Silvio Savarese. 2022a. igibson 2.0: Object-centric simulation for robot learning of everyday household tasks. In *Proceedings of the 5th Conference on Robot Learning*, volume 164 of *Proceedings of Machine Learning Research*, pages 455–465. PMLR.

Chengshu Li, Ruohan Zhang, Josiah Wong, Cem Gokmen, Sanjana Srivastava, Roberto Martín-Martín, Chen Wang, Gabrael Levine, Michael Lingelbach, Jiankai Sun, Mona Anvari, Minjune Hwang, Manasi Sharma, Arman Aydin, Dhruva Bansal, Samuel Hunter, Kyu-Young Kim, Alan Lou, Caleb R. Matthews, and 9 others. 2022b. BEHAVIOR-1K: A benchmark for embodied AI with 1, 000 everyday activities and realistic simulation. In *CoRL*, volume 205 of *Proceedings of Machine Learning Research*, pages 80–93. PMLR.

Wei Li, Renshan Zhang, Rui Shao, Jie He, and Liqiang Nie. 2025. Cogvla: Cognition-aligned vision-language-action model via instruction-driven routing & sparsification. *Advances in neural information processing systems*.

Xuanlin Li, Kyle Hsu, Jiayuan Gu, Oier Mees, Karl Pertsch, Homer Rich Walke, Chuyuan Fu, Ishikaa Lunawat, Isabel Sieh, Sean Kirmani, Sergey Levine, Jiajun Wu, Chelsea Finn, Hao Su, Quan Vuong, and Ted Xiao. 2024. Evaluating real-world robot manipulation policies in simulation. In *CoRL*, volume 270 of *Proceedings of Machine Learning Research*, pages 3705–3728. PMLR.

Zechang Li, Yuxuan Lai, Yansong Feng, and Dongyan Zhao. 2020. Domain adaptation for semantic parsing. In *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence, IJCAI 2020*, pages 3723–3729. ijcai.org.

Zhixuan Liang, Yizhuo Li, Tianshuo Yang, Chengyue Wu, Sitong Mao, Liuao Pei, Xiaokang Yang, Jiangmiao Pang, Yao Mu, and Ping Luo. 2025. Discrete diffusion vla: Bringing discrete diffusion to action decoding in vision-language-action policies. *arXiv preprint arXiv:2508.20072*.

Nir Lipovetzky and Hector Geffner. 2012. Width and serialization of classical planning problems. In *ECAI*, volume 242 of *Frontiers in Artificial Intelligence and Applications*, pages 540–545. IOS Press.

Nir Lipovetzky and Hector Geffner. 2017. Best-first width search: Exploration and exploitation in classical planning. In *AAAI*, pages 3590–3596. AAAI Press.

Yueen Ma, Zixing Song, Yuzheng Zhuang, Jianye Hao, and Irwin King. 2025. A survey on vision-language-action models for embodied ai. *Preprint*, arXiv:2405.14093.

Vivek Myers, Chunyuan Zheng, Oier Mees, Kuan Fang, and Sergey Levine. 2024. Policy adaptation via language optimization: Decomposing tasks for few-shot imitation. In *CoRL*, volume 270 of *Proceedings of Machine Learning Research*, pages 1402–1426. PMLR.

Abby O'Neill, Abdul Rehman, Abhiram Maddukuri, Abhishek Gupta, Abhishek Padalkar, Abraham Lee, Acorn Pooley, Agrim Gupta, Ajay Mandlekar, Ajinkya Jain, Albert Tung, Alex Bewley, Alexander Herzog, Alex Irpan, Alexander Khazatsky, Anant Rai, Anchit Gupta, Andrew E. Wang, Anikait Singh, and 260 others. 2024. Open x-embodiment: Robotic learning datasets and RT-X models : Open x-embodiment collaboration. In *ICRA*, pages 6892–6903. IEEE.

Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy V. Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mido Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, and 7 others. 2024. Dinov2: Learning robust visual features without supervision. *Trans. Mach. Learn. Res.*, 2024.

Shishir G. Patil, Tianjun Zhang, Xin Wang, and Joseph E. Gonzalez. 2024. Gorilla: Large language model connected with massive apis. In *NeurIPS*.

Yujia Qin, Shihao Liang, Yining Ye, Kunlun Zhu, Lan Yan, Yaxi Lu, Yankai Lin, Xin Cong, Xiangru Tang, Bill Qian, Sihan Zhao, Lauren Hong, Runchu Tian, Ruobing Xie, Jie Zhou, Mark Gerstein, Dahai Li, Zhiyuan Liu, and Maosong Sun. 2024. Toolllm: Facilitating large language models to master 16000+ real-world apis. In *ICLR*. OpenReview.net.

Scott E. Reed, Konrad Zolna, Emilio Parisotto, Sergio Gómez Colmenarejo, Alexander Novikov, Gabriel Barth-Maron, Mai Gimenez, Yury Sulsky, Jackie Kay, Jost Tobias Springenberg, Tom Eccles, Jake Bruce, Ali Razavi, Ashley Edwards, Nicolas Heess, Yutian Chen, Raia Hadsell, Oriol Vinyals, Mahyar Bordbar, and Nando de Freitas. 2022. A generalist agent. *Trans. Mach. Learn. Res.*, 2022.

Lucy Xiaoyang Shi, Brian Ichter, Michael Robert Equi, Liyiming Ke, Karl Pertsch, Quan Vuong, James Tanner, Anna Walling, Haohuan Wang, Niccolo Fusai, Adrian Li-Bell, Danny Driess, Lachy Groom, Sergey Levine, and Chelsea Finn. 2025. Hi robot: Open-ended instruction following with hierarchical vision-language-action models. In *ICML*.

Mohit Shridhar, Jesse Thomason, Daniel Gordon, Yonatan Bisk, Winson Han, Roozbeh Mottaghi, Luke Zettlemoyer, and Dieter Fox. 2020. Alfred: A benchmark for interpreting grounded instructions for everyday tasks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10740–10749.

Mustafa Shukor, Dana Aubakirova, Francesco Capuano, Pepijn Kooijmans, Steven Palma, Adil Zouitine, Michel Aractingi, Caroline Pascal, Martino Russi, Andres Marafioti, and 1 others. 2025. Smolvla: A vision-language-action model for affordable and efficient robotics. *arXiv preprint arXiv:2506.01844*.

Wenxuan Song, Jiayi Chen, Pengxiang Ding, Han Zhao, Wei Zhao, Zhide Zhong, Zongyuan Ge, Zhijun Li, Donglin Wang, Lujia Wang, Jun Ma, and Haoang Li. 2025. PD-VLA: accelerating vision-language-action model integrated with action chunking via parallel decoding. In *IROS*, pages 13162–13169. IEEE.

Gerald Jay Sussman. 1975. *A computer model of skill acquisition*. Elsevier Science Inc.

Hongyu Wang, Chuyan Xiong, Ruiping Wang, and Xilin Chen. 2025. Bitvla: 1-bit vision-language-action models for robotics manipulation. *arXiv preprint arXiv:2506.07530*.

Jeannette M. Wing. 2010. Computational thinking: What and why? *The Link*.

Lei Xiao, Jifeng Li, Juntao Gao, Feiyang Ye, Yan Jin, Jingjing Qian, Jing Zhang, Yong Wu, and Xiaoyuan Yu. 2025. Ava-vla: Improving vision-language-action models with active visual attention. *arXiv preprint arXiv:2511.18960*.

Youguang Xing, Xu Luo, Junlin Xie, Lianli Gao, Heng Tao Shen, and Jingkuan Song. 2025. Shortcut learning in generalist robot policies: The role of dataset diversity and fragmentation. In *9th Annual Conference on Robot Learning*.

Weimin Xiong, Yifan Song, Qingxiu Dong, Bingchan Zhao, Feifan Song, XWang, and Sujian Li. 2025. MPO: Boosting LLM agents with meta plan optimization. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 3914–3935, Suzhou, China. Association for Computational Linguistics.

Ruochu Yang, Fumin Zhang, and Mengxue Hou. 2024. Oceanplan: Hierarchical planning and replanning for natural language AUV piloting in large-scale unexplored ocean environments. In *WUWNet*, pages 11:1–11:5. ACM.

Shuai Yang, Hao Li, Yilun Chen, Bin Wang, Yang Tian, Tai Wang, Hanqing Wang, Feng Zhao, Yiyi Liao, and Jiangmiao Pang. 2025. Instructvla: Vision-language-action instruction tuning from understanding to manipulation. *Preprint*, arXiv:2507.17520.

Cheng Yin, Yankai Lin, Wang Xu, Sikyuen Tam, Xiangrui Zeng, Zhiyuan Liu, and Zhouping Yin. 2025. Deepthinkvla: Enhancing reasoning capability of vision-language-action models. *arXiv preprint arXiv:2511.15669*.

Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. 2023. Sigmoid loss for language image pre-training. In *ICCV*, pages 11941–11952. IEEE.

Huaixiu Steven Zheng, Swaroop Mishra, Xinyun Chen, Heng-Tze Cheng, Ed H. Chi, Quoc V. Le, and Denny Zhou. 2024. Take a step back: Evoking reasoning via abstraction in large language models. In *ICLR*.

Yifan Zhong, Fengshuo Bai, Shaofei Cai, Xuchuan Huang, Zhang Chen, Xiaowei Zhang, Yuanfei Wang, Shaoyang Guo, Tianrui Guan, Ka Nam Lui, and 1 others. 2025. A survey on vision-language-action models: An action tokenization perspective. *arXiv preprint arXiv:2507.01925*.

Brianna Zitkovich, Tianhe Yu, Sichun Xu, Peng Xu, Ted Xiao, Fei Xia, Jialin Wu, Paul Wohlhart, Stefan Welker, Ayzaan Wahid, Quan Vuong, Vincent Vanhoucke, Huong T. Tran, Radu Soricut, Anikait Singh, Jaspiar Singh, Pierre Sermanet, Pannag R.

Sanketi, Grecia Salazar, and 35 others. 2023. RT-2: vision-language-action models transfer web knowledge to robotic control. In *CoRL*, volume 229 of *Proceedings of Machine Learning Research*, pages 2165–2183. PMLR.

A Technical Appendix

Appendix Table of Contents

B Anticipatory defense on the choice of test environments and metrics

In this work, we selected Mini-BEHAVIOR as our testbed. This choice is motivated by several key considerations aligned with our research objectives. Our research objective is not to solve the full stack of robotic embodiment (which includes perception, continuous control, and physics), but specifically to isolate and analyze the impact of instruction granularity on language grounding and long-horizon planning.

Therefore, high-fidelity simulators (e.g., iGibson (Li et al., 2022a), BEHAVIOR-1K (Li et al., 2022b)) introduce significant noise through complex texture rendering and continuous motor control. While realistic, these factors act as confounders. When an agent fails in such environments, it is often difficult to attribute the failure to a lack of linguistic understanding (granularity issues) or a failure in low-level execution (e.g., grasping physics or occlusion). Mini-BEHAVIOR retains the high-level decision complexity of household tasks—long horizons, multi-object interactions, and state dependencies—while abstracting away low-level sensorimotor noise.

In the perspective of fast prototyping and iterative experimentation, we have to also consider the rendering and physics simulation costs. Mini-BEHAVIOR's grid-based discrete control and simplified visual representation allow for rapid data generation and model training.

Another consideration is the diversity of the task structures so that we can systematically vary instruction granularity across a wide range of scenarios. As such, Crafter (Hafner, 2021), which focuses on survival in a single domain, lacks the diverse task structures needed to obtain robust insights about instruction granularity.

Table 3 provides a detailed comparison of Mini-BEHAVIOR against other prominent benchmarks, highlighting why they were less suitable for this specific investigation.

C Symbolic Representation of the Embodied Task

We formalize the Mini-BEHAVIOR environment as a deterministic grid-based symbolic world. This allows us to represent the challenge as a classical planning problem $\mathcal{P} = \langle \mathcal{F}, \mathcal{A}, s_0, G \rangle$, where:

- $\mathcal{F}$ is a finite set of propositional fluents representing the state variables of the environment (e.g., object locations, states like *cooked* or *open*).
- $\mathcal{A}$ is a finite set of actions available to the agent.
- $s_0 \subseteq \mathcal{F}$ is the initial state configuration.
- $G \subseteq \mathcal{F}$ is the goal condition that must be satisfied.

A state $s \in \mathcal{S}$ is defined as a truth valuation over $\mathcal{F}$, with the state space $\mathcal{S} \subseteq 2^{\mathcal{F}}$ encompassing all valid combinations of fluent truth values.